

CS 253: Data Structures and Algorithms

Randomized Final Study Problem

Sequence-Based Structures

Red-Black Tree Tracing Challenge

Start from an empty red-black tree. Insert the keys in [77, 13, 37, 20, 68, 62, 65, 88, 53, 31]. Then remove the keys in [88, 77], showing every recoloring and rotation needed to restore the red-black invariants.

Instructor Reference

Final red-black tree after all updates. Insert rotations: 4, insert recolors: 14, delete rotations: 1, delete recolors: 3.

